ECE 565 - DATA MINING

Weekly Progress Report Template

Deadline: Every Tuesday by 11:59 PM

Spring 2026

Instructor: Asst. Prof. Dr. Asli Eyecioglu Ozmutlu

INSTRUCTIONS

1. Fill out this report every week with your project progress.
2. Submit as PDF (StudentID_Surname_Week#.pdf) on Canvas by Tuesday 11:59 PM.
3. Include actual results (numbers, tables, figures) — not just descriptions of what you plan to do.
4. Compare your results against the papers from your proposal. Use the comparison table provided.
5. Be honest about problems. If something didn't work, explain why and what you tried.
6. **The 'Next Week Plan' section is your commitment — I will check it against the following week's report.**

WEEKLY PROGRESS REPORT

Student Name:	Resul Ozkale	**Student ID:**	2540041007
Week Number:	Week 13 - Validation Protocol Repair and Final Attempt Stress Testing	**Date:**	May 12, 2026
Project Title:	SCRE-Credit: Stability- and Cost-Regularized Ensemble for Credit Risk		
Dataset:	Primary: UCI Default of Credit Card Clients / Taiwan. External validation: FICO HELOC. German Credit / Option A is archived and out of active scope.		

SECTION 1: Work Completed This Week

Describe concretely what you accomplished this week. Use the checklist below and provide details for each completed item.

☐	Task	Details / Evidence
Done	Data loading & exploration	No scope change this week. Taiwan remains the primary dataset and HELOC remains the external validation dataset. German Credit / Option A remains archived and out of active scope.
Done	Data cleaning	No new dataset was added and no target-derived feature was created. The cleaned Taiwan and HELOC datasets from the reproducible pipeline remain the basis for all experiments.
Done	Preprocessing implementation	The V2 protocol protects the test split: model/policy selection is validation-only; test is used only after locking. No split-before-resampling issue was found in the final audit.
Done	Feature engineering review	Existing row-wise Taiwan and HELOC features were kept frozen. The final-attempt experiments did not introduce new features, so the earlier leakage controls remain intact.
Done	Model training / model-policy evaluation	Evaluated V2 locked policies, strict literature reproduction, advanced imbalance boosting, deep tabular models, interpretable middle models, capacity-aware review, and decision curves.

Done	Validation protocol repair	Fixed the previous NOT READY problem by creating selection_protocol_v2, validation-only candidate registries, locked final policies, and separate held-out test evidence tables.
Done	Manual-review cost model	Separated binary FN/FP expected cost from review-adjusted operational cost. Manual-review workload and review-cost assumptions are now explicitly documented.
Done	False-positive reduction	Completed precision-constrained thresholds, cost-matrix sensitivity, manual-review band optimization, eligible model selection, and false-positive profile analysis.
Done	Literature / advanced reproduction	Tested SMOTE, BorderlineSMOTE, SMOTE-Tomek, KMeansSMOTE, class weighting, MLP/BP neural networks, EBM, and monotonic models under leakage-free protocol.
Partial	Final-attempt audit	No test-selection leakage was detected, but final locked-test manual-review rows failed metric-consistency identities. V2 remains report-ready; final attempt needs repair before replacing V2.

SECTION 1b: Show Your Work

Paste or attach at least one piece of evidence from this week: a code output, a plot, a table of results, a screenshot, or a confusion matrix. Do NOT submit a report with only text and no evidence.

Evidence	Model / Policy	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	PR-AUC	Cost	Comment
Taiwan V2 locked policy	Best CatBoost manual-review band	0.793	0.529	0.575	0.854	0.551	0.785	0.564	2705.500	Report-ready baseline.
HELOC V2 locked policy	Best Scorecard manual-review band	0.716	0.683	0.846	0.574	0.756	0.801	0.804	764.500	Report-ready external-validation baseline.
Taiwan final attempt	XGBoost manual-review	0.761	0.499	0.597	0.830	0.544	0.780	0.559	2775.500	Diagnostic only; audit blocked metric

										consistency.
HELOC final attempt	Scorecard manual-review	0.716	0.683	0.846	0.574	0.756	0.801	0.804	764.500	Diagnostic only; same policy family as V2.

Important transparency note: V2 is currently the report-ready result. The later final-attempt package is useful and methodologically informative, but its final locked-test manual-review table is blocked by metric-consistency audit failures. No test-selection leakage was detected

You can access all my results and code via my GitHub link; godsoftware/Data_Mining_Project.

Additional Evidence: Old Aggressive Threshold vs V2 Manual-Review Policy

Dataset / system	Policy	Precision	Recall	Specificity	FP	FN	Cost	Comment
Taiwan old aggressive CatBoost	Cost-threshold binary policy	0.351	0.809	0.576	1982	253	3247	High recall but too many false positives.
Taiwan V2 CatBoost	Manual-review band	0.529	0.575	0.854	680	564	2705.500	FP reduced by 1302 vs old CatBoost; precision and specificity improved.
HELOC old aggressive CatBoost reference	Cost-threshold binary policy	0.558	0.981	0.157	798	20	898	Very high recall but low specificity.
HELOC V2 Scorecard	Manual-review band	0.683	0.846	0.574	403	158	764.500	FP reduced by 395 vs old reference; better operational balance.

Additional Evidence: Final-Attempt Experiment Summary

Experiment	Best validation config / model	PR-AUC	ROC-AUC	Precision	Recall	Specificity	Cost	Interpretation
Strict literature reproduction - Taiwan	XGBoost + no resampling	0.564	0.789	0.375	0.775	0.633	3209.000	No reliable V2 replacement.
Strict literature reproduction - HELOC	CatBoost Balanced	0.799	0.803	0.574	0.975	0.215	873.000	High literature-style jump not reproduced.
Advanced imbalance - Taiwan	XGBoost xgboost_scale_pos_weight_5	0.562	0.788	0.514	0.588	0.842	2648.500	Useful validation candidate; test evidence did not beat V2.
Advanced imbalance - HELOC	CatBoost catboost_baseline	0.783	0.801	0.672	0.870	0.540	747.000	Scorecard remains safer for HELOC.
Deep tabular - Taiwan	mlp_256-128-64_relu_base	0.552	0.780	0.687	0.354	0.954	4499.000	DNN did not deliver better operational policy.
Deep tabular - HELOC	mlp_128m64_l2_0.001	0.811	0.804	0.730	0.762	0.694	1510.000	Raw PR-AUC useful, but not final operational winner.
Interpretable - Taiwan	Monotonic LightGBM Monotonic LightGBM conservative	0.562	0.788	0.508	0.585	0.839	2668.000	Governance-friendly alternative.

Interpretable - HELOC	EBM EBM interactions=10	0.786	0.801	0.701	0.839	0.611	726.000	Scorecard/EBM remain strong interpretable baselines.

Additional Evidence: Capacity-Aware Manual Review

Dataset	Best validation ranking model	Capture@10%	Capture@20%	Capture@30%	Precision@20%	Lift@20%	Comment
Taiwan	SCRE-Optimized probability ranking	0.321	0.529	0.647	0.585	2.645	Good framing for review prioritization.
HELOC	SCRE-Optimized probability ranking	0.182	0.340	0.500	0.884	1.698	Useful external validation for ranking.

Additional Evidence: Decision Curve / Net Benefit

Dataset	Best validation model	Useful threshold range	Max net benefit	Max NB threshold	Comment
Taiwan	SCRE-Pareto probability ranking	0.01-0.80	0.213	0.010	Supports screening/manual-review use.
HELOC	Best CatBoost probability ranking	0.03-0.80	0.515	0.010	Models provide benefit over naive strategies in useful ranges.

Additional Evidence: Final Multi-Track Selection

Dataset	Track	Selected system	Reason / note
Taiwan	Best raw classifier	XGBoost xgboost_spw_5_mds_1	Highest validation PR-AUC after tie-breaking

			by ROC-AUC, F1, and calibration. No test metric was consulted.
Taiwan	Best operational binary policy	XGBoost xgboost_scale_pos_weight_5	Lowest validation binary expected cost among eligible binary policies, with PR-AUC/ECE/F1 used only as validation tie-breakers.
Taiwan	Best manual-review policy	XGBoost xgboost_scale_pos_weight_5	Lowest validation review-adjusted cost under review_cost=0.5 among manual-review candidates satisfying MR-rate and high-risk quality constraints.
Taiwan	Best capacity-aware reviewer	SCRE-Optimized probability ranking	Highest validation default capture@20%, then precision@20%, lift@20%, and net value@20%.
Taiwan	Best interpretable model	Monotonic LightGBM conservative	Best validation-governed interpretable candidate after enforcing manual-review feasibility; scorecard is preferred when it remains competitive and feasible.
Taiwan	Best research framework	SCRE-Optimized	SCRE variant selected for reliability integration, validation AUC/PR-AUC, and capacity-review support. This is a framework track, not a claim of universal predictive dominance.
Taiwan	Final recommended system	Operational screening: XGBoost xgboost_scale_pos_weight_5 (manual_review_band_mr_le_0_30); review prioritization: SCRE-Optimized probability ranking; interpretable benchmark: Monotonic LightGBM conservative; research framework: SCRE-Optimized.	Taiwan changes partially from V2: the validation-only manual-review winner shifts marginally from V2 CatBoost to advanced XGBoost (review-adjusted cost 2648.5), while SCRE-Optimized

			is kept for top-k review prioritization and Monotonic LightGBM is the strongest interpretable middle model.
HELOC	Best raw classifier	MLP/BP Neural Network mlp_128m64_l2_0.001	Highest validation PR-AUC after tie-breaking by ROC-AUC, F1, and calibration. No test metric was consulted.
HELOC	Best operational binary policy	WOE scorecard unweighted	Lowest validation binary expected cost among eligible binary policies, with PR-AUC/ECE/F1 used only as validation tie-breakers.
HELOC	Best manual-review policy	Best Scorecard	Lowest validation review-adjusted cost under review_cost=0.5 among manual-review candidates satisfying MR-rate and high-risk quality constraints.
HELOC	Best capacity-aware reviewer	SCRE-Optimized probability ranking	Highest validation default capture@20%, then precision@20%, lift@20%, and net value@20%.
HELOC	Best interpretable model	WOE scorecard unweighted	Best validation-governed interpretable candidate after enforcing manual-review feasibility; scorecard is preferred when it remains competitive and feasible.
HELOC	Best research framework	SCRE-Optimized	SCRE variant selected for reliability integration, validation AUC/PR-AUC, and capacity-review support. This is a framework track, not a claim of universal predictive dominance.

HELOC	Final recommended system	Operational screening: Best Scorecard (manual_review_band); review prioritization: SCRE-Optimized probability ranking; interpretable benchmark: WOE scorecard unweighted; research framework: SCRE-Optimized.	HELOC operational selection remains close to V2: Scorecard-style manual review remains the cleanest validation-only operational policy, while SCRE-Optimized ranks best for capacity@20 review prioritization.

SECTION 2: Preliminary Results & Literature Comparison

Report your current numerical results and compare them against the papers from your proposal. Fill in the table below.

Model / Paper	Method	Accuracy	F1-Score	AUC-ROC	Other Metric	Best?	Notes
Alam et al. (2020)	GBDT + K-means SMOTE on Taiwan	~0.887	Not reported	Not reported	Reported high accuracy	Not matched	Week 13 leakage-free reproduction did not show a comparable jump.
K-means SMOTE + BP NN (2021)	Class imbalance handling + neural net	Not reported	Not reported	0.929	AUC improved 0.765 to 0.929	Not matched	KMeansSMOTE and MLP/BP reproduction did not produce a reliable operational breakthrough.
Interpretable DL + XAI (2023)	Deep model with XAI	0.835	Not reported	Not reported	Sensitivity 0.882; specificity 0.988	Not claimed	Current project remains more conservative and audit-focused.
DNN empirical study (2025)	Deep neural network on Taiwan	0.818	0.482	0.770	G-mean 0.603	Partial context	Comparable to some raw metrics, but not enough to justify automatic rejection.
Week 12 old	Cost-sensitive	0.627	0.490	0.785	Precision 0.351; recall	Problem baseline	High recall but excessive false positives.

aggressive CatBoost	binary threshold				0.809; PR-AUC 0.564; cost 3247		
Week 13 V2 Taiwan	CatBoost manual-review band	0.793	0.551	0.785	PR-AUC 0.564; precision 0.529; specificity 0.854; review-adjusted cost 2705.500	Best safe Taiwan baseline	Report-ready and more operationally defensible.
Week 13 V2 HELOC	Scorecard manual-review band	0.716	0.756	0.801	PR-AUC 0.804; precision 0.683; specificity 0.574; review-adjusted cost 764.500	Best safe HELOC baseline	Interpretable and externally useful.
SCRE-Credit role	Reliability-aware framework / ranking component	0.645	0.492	0.785	SCRE-Optimized test PR-AUC 0.567; cost 3312 under FN=5/FP=1	Framework, not universal winner	Useful for reliability-aware comparison and ranking, not a dominance claim.

Note: Literature rows use “Not reported” only when that exact metric was not reported in the referenced proposal source. Project rows use actual values from generated output tables.

SECTION 2b: Results Discussion

Q1: Compared with Week 12, the most important improvement is methodological. Final model and policy selection is now validation-only, and held-out test results are reported only after policy locking.

Q2: The Week 12 problem was that the cost-optimal CatBoost threshold was too aggressive: high recall, low precision, and many false positives. Week 13 addressed this by moving toward constrained thresholding and manual-review policies.

Q3: The V2 Taiwan policy is more operationally defensible than the old cost-threshold policy because precision increased from 0.351 to 0.529, specificity increased from 0.576 to 0.854, and false positives fell from 1982 to 680.

Q4: Strict literature reproduction did not produce the very high scores reported in some papers when resampling was kept inside training folds and the test distribution remained natural. This supports a careful, leakage-aware discussion rather than a claim of state-of-the-art accuracy.

Q5: SCRE-Credit should be retained as a reliability-aware framework and review-ranking component. It should not be presented as a model that beats CatBoost, Scorecard, XGBoost, and LightGBM in every metric.

SECTION 3: Challenges & Problems Encountered

Describe any problems you faced this week. Be specific about error messages, unexpected results, or conceptual difficulties.

Problem	What I Tried / Found	Status (Solved / Ongoing)
Final-attempt audit BLOCKED	The audit found 9 critical metric-consistency failures in final locked-test manual-review rows. Recall, specificity, and F1 were not always recomputable from the displayed confusion counts.	Ongoing - repair needed before final-attempt tables can replace V2.
V2 vs final-attempt decision	Final-attempt Taiwan XGBoost improved recall slightly but worsened precision, specificity, FP count, and review-adjusted cost compared with V2.	Ongoing - V2 remains the safer report baseline.
Manual-review denominator complexity	Manual-review policies mix full-population metrics, auto-decision-only metrics, and review workload cost. These must be separated in final tables.	Ongoing - next technical repair.
Automatic rejection risk	Precision and false-positive behavior are not strong enough for direct automatic credit rejection.	Solved by safe screening/manual-review wording.
Literature score gap	Leakage-free reproduction did not match very high published scores. Likely reasons include protocol differences, resampled test distributions, threshold choices, or weaker leakage controls in some comparisons.	Solved as methodological discussion.

SECTION 4: Plan for Next Week

List 3-5 specific, measurable tasks you will complete by next Tuesday. These will be checked against your next report.

#	Planned Task (be specific)	Expected Deliverable
1	Repair final-attempt locked-test manual-review metric tables.	Separate full-population metrics, auto-decision metrics, manual-review counts, review workload cost, and binary diagnostic cost.
2	Re-run final-attempt brutal audit after the table repair.	Updated final_attempt_audit.md showing whether the BLOCKED status is resolved.

3	Choose final report baseline.	Use V2 as the main report result unless the repaired final attempt clearly passes audit and improves operational balance.
4	Draft final project report sections.	Methods, validation protocol, datasets, model comparison, manual-review policy, capacity analysis, limitations, and safe claims.
5	Prepare final figures and appendix tables.	Manual-review workflow, capacity@K/lift curves, decision curve, V2 locked results, claim-control matrix, and audit summary.

SECTION 5: Self-Assessment

Question	Your Answer
Overall project completion (%)	Approximately 90% if V2 is used as the final report baseline; approximately 80-85% if the final-attempt package must become the main result.
Did you complete last week's planned tasks? (Yes / Partially / No)	Yes. Precision-constrained thresholds, cost sensitivity, decision curve analysis, training-level imbalance testing, and manual-review optimization were all implemented.
If not, why?	The planned experiments were completed. The remaining issue is not missing experimentation; it is final-attempt metric-consistency cleanup before those results can replace V2.
Are you on track to finish the project on time?	Yes, if the final report uses V2 as the report-ready baseline and the final attempt is either repaired or presented as a stress-test appendix.
Do you need help from the instructor? If yes, on what?	Potentially yes: feedback on whether the report should emphasize the V2 READY policy or include the final-attempt stress tests as a secondary appendix.

SUGGESTED WEEKLY MILESTONES

Use this as a guide for pacing your project. We assume you have already done 1-5. Your weekly reports should roughly follow this timeline.

Week	Focus Area	Expected Tasks	Current status in this project
1	Data Loading & EDA	Load dataset, check shape/types/missing values, compute basic statistics, create distribution plots	Completed earlier

2	Data Cleaning & Preprocessing	Handle missing values, duplicates, category cleaning, scaling, train-only preprocessing	Completed earlier
3	Feature Engineering	Create domain-specific features and final feature set	Completed earlier and frozen
4	Baseline Model	Train LR/RF/boosting/scorecard baselines	Completed earlier
5	Imbalance Handling & Ablation	Compare sampling, cost-sensitive, and ablation variants	Completed and extended this week
6	Proposed Model Implementation	Implement SCRE-Credit and compare with baselines	Completed earlier; role refined this week
7	Hyperparameter Optimization	Run Optuna / tuning and compare before/after	Completed earlier
8	Explainability & Analysis	Run SHAP/LIME, stability, faithfulness, subgroup checks	Completed earlier; stability claims controlled
9	Final Evaluation & Writing	Final hold-out evaluation, audit, figures, report writing	Current focus
10	Final Report Submission	Write discussion, conclusion, limitations, future work	Next step after final metric cleanup

WEEKLY REPORT GRADING CRITERIA

Criterion	Weight	How this report addresses it
Progress Evidence	30%	Concrete artifacts, tables, figures, and audit results are included.
Results & Comparison	30%	Numerical results are compared with proposal literature and current V2/final-attempt evidence.
Critical Thinking	20%	The report explains why V2 remains safer and why final attempt is blocked.
Planning & Commitment	20%	Next-week tasks are specific and measurable.

IMPORTANT WARNINGS

- Reports with NO numerical results or evidence will receive 0 for 'Progress Evidence'.
- Submitting the same content two weeks in a row will receive 0 for 'Progress Evidence'.
- If results exceed the literature by a suspiciously large margin, you must explain why. Unexplained perfect scores will be flagged.
- AI-generated text is not a substitute for your own analysis. Show YOUR understanding of YOUR results.
- Late submissions: same policy as the project proposal (up to 1 hour = 25% penalty, after 1 hour = not accepted).